

CS608 – WEB DEVELOPMENT LAB
M TECH
Exercise 10

Develop a Web-Based Real-Time Chat System using Socket Programming in PHP

Requirements:

1. Create a main page (index.php) with 3 options (simple links or buttons):
 - a. Start Chat Server
 - b. Join Chat Room
 - c. View Active Users
 - d. Logout (redirects to index.php and clears session if used)
2. Create 'Start Chat Server' section:
 - a. Implement a PHP socket server to listen for incoming connections.
 - b. The server should:
 - i. Accept multiple client connections.
 - ii. Maintain a list of connected users.
 - iii. Broadcast messages to all clients.
 - c. Use PHP socket functions (socket_create, socket_bind, socket_listen, socket_accept).
3. Create 'Join Chat Room' page:
 - a. Provide a form with Username and Message input.
 - b. Establish socket connection to the server.
 - c. Send messages using PHP/AJAX.
 - d. Display incoming messages dynamically without refreshing.
4. Display chat messages dynamically:
 - a. Use AJAX or long polling.
 - b. Show messages in real-time chat window.
 - c. Format: Username: Message
 - d. If no messages: 'No messages yet. Start the conversation!'
5. Disconnect functionality:
 - a. Provide 'Disconnect' button.
 - b. Remove user from server list.
 - c. Notify others when a user leaves.
6. Search messages:
 - a. Add search box.
 - b. Filter messages dynamically using AJAX.
7. View Active Users page:
 - a. Show total number of users.
 - b. List usernames.
 - c. Refresh button using AJAX.